DecoLeaves: Exploring NFC technology for Energy Harvesting and Concealed Input on Interactive Fashion with Dynamic Graphics

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Figure 1: DecoLeaves Shirt with embedded electrochromic displays on the shoulder. (a) front view of the shirt; (b) two visual states of the leaves; (c) user putting a smart phone to switch the leaves' visual state.

ABSTRACT

We present the design and prototype of the DecoLeaves shirt, a battery-free wraparound style shirt that gives the autonomy to its wearer for modifying her appearance by moving a smartphone secretly in the shirt's pocket. It is designed to express emotions, status or the wearer's self in general, without being too over the top and can be worn on casual occasions. To achieve its functionality, it utilizes two NFC-based wireless energy harvesting components inside to pocket as a secret input technique to switch leaf-shaped electrochromic (EC) display embellishments on the shoulder without needing an embedded battery on the shirt. Our work contributes the utilization of NFC technology as an energy harvesting technique, as well as a secret input modality for interactive fashion to alter the visual state of the garment.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing**; **Interaction paradigms**; **Interaction design**; *Displays and imagers.*

KEYWORDS

interactive fashion, social wearables, energy harvesting, battery-free, electrochromic displays, concealed interface, NFC

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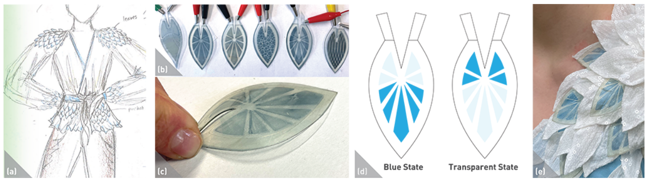

Figure 2: Design process. (a) initial sketch of the shirt; (b) alternative graphic designs; (b) the bent shape of the leaf displays; (c) blue and transparent states of the final display design; (d) sequin fabric blocking the ink on the transparent state.

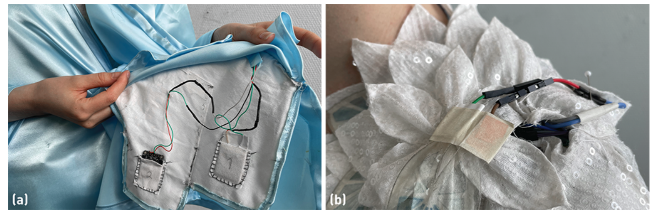

Figure 3: Placement of the electronics. (a) NFC boards placed in the pocket's compartments; (b) header connections of the displays

1 INTRODUCTION

Nowadays, interactive devices are criticized as they cause an immense amount of battery waste that is hard to recycle [14]. Interactive fashion devices are no exception to this problem as they often emphasize aesthetic approaches using embedded batteries to power output modalities such as thermochromic materials [10] or LEDs [1]. Considering the usability issues regarding batteries on wearables (i.e., charging or replacing [8]) are also becoming important factors for future wearables, exploring wearables without embedded batteries deserves further exploration.

Motivated by this challenge, in this work, we focus on Near Field Communication (NFC) technologies to enable visual changes on interactive fashion pieces without an embedded battery. With NFC, it is possible to harvest energy wirelessly from other electronic devices, i.e. smartphones, to power interactive systems that require low power consumption. Such systems are demonstrated by prior wearable studies, i.e., [4, 5]. These examples show, for instance, how putting a smartphone in the pocket of a smart garment can activate visual changes. In our exploration, different from prior examples, we focus on how NFC can be utilized in an interactive fashion design that enables a secret input from the wearer to change the visual state of the garment. Secret input can be valuable to influence interpersonal interactions without input being noticeable by others [2]. Examples of these cases might be secretly letting close ones know that the wearer is in an emergency situation or expressing the mood or status subtly within a conversation. While such concealed interfaces are explored in wearables by using natural gestures accomplished on the body (i.e., playing with the hair [13]) or on the garments such as detecting touch patterns on the fabric of the pocket [9], how to combine these approaches within battery-free interactive fashion designs is open for exploration.

In the presented work, by being inspired by the NFC technologies' potential to be used as a concealed interface without the need for an embedded battery, we designed and implemented DecoLeaves (Figure 1). It is a wraparound style shirt that gives the autonomy to its wearer for modifying her appearance implicitly by moving a smartphone in the shirt's pocket. This exhibition paper presents the design process and implementation of the shirt.

2 THE CONCEPT

The Decoleaves is a shirt that allows users to switch the visual state of the leaf-shaped embellishments on the shoulder without the need for a battery embedded in the garment. The shirt gives the wearer power to secretly alter the visual appearance of the garment to express their feelings, status, or in a broader sense, themselves in social situations. The wearer can move the phone closer to either one of the two predefined positions inside the pocket for turning the leaves blue or transparent. It is designed to subtly influence social interactions without being too over the top and is designed to be suitable for everyday use as well as for parties and special occasions. Also, thanks to the wrap-around style of this shirt, it is suitable for many users of different sizes.

2.1 Design Process and Implementation

The design of the jacket started with the creation of sketches exploring how electrochromic (EC) displays can be embedded in an interactive fashion. These displays are good candidates for battery-free interactive fashion as they require low power to be switched between two pre-defined graphical states, and can be fabricated free-form, transparent and flexible [6]. They are also lightweight, which eases their placement on the body, as heavy parts would limit the garment's design [15]. We focused on a concept with EC displays placed the shoulder as embellishment (Figure 2a) because the shoulder is closer to the head and can allow the visual change to be more distinguishable from observers.

After selecting the concept, we further explored visuals for the displays by being inspired by the surface of leaves (Figure 2a). As we aimed to keep the graphics simple to draw attention to several leaves instead of one, we selected the design where the graphic is a stylized version of the leaf's pattern (Figure 2c).Alongside these displays, in the final design, we integrated leaf-shaped white woven sequin fabric details to provide a background for displays, as well as using them to conceal the blue ink in the transparent state (Figure 2c/d). The shape of the displays was designed with a cut to be combined in the stem part, similar to darts in garment design [11]. This allowed them to be bent convexly and transformed into a three-dimensional form (Figure 2b). In the final implementation, light blue satin (100% polyester) was selected as the base fabric of the shirt, which matches well with and emphasizes the blue ink of the displays.

For using NFC technology as a concealed interface on the shirt, we focused on the secret interactions that can occur within the pocket. The pocket is designed to enable a smartphone to power and change the visual states of the displays without needing to take out the phone. For this, two NFC-based energy harvesting boards (Adafruit ST25DV16K I2C RFID EEPROM Breakout) are placed in two different compartments of the pocket (Figure 3-a). Each one of these boards is capable of harvesting energy (approx. 1.5V) when an NFC-enabled phone is in close proximity. While one of these boards provides a positive voltage to the display to turn the blue state on, the other one sends a negative voltage to enable the transparent state. The electrical connections between the boards and the displays were made possible with four flexible wires (GND and 1.5V from each board). The wires were routed from the seam of the shirt from the boards to the shoulder (Figure 3-a). At the shoulder end of the routes, we soldered the opposite wires of the boards (GND from one board to 1.5V from the other) together for connecting them easily to the displays. The resulting two connections then were multiplied by using more wires on the shoulder hidden under the fabric leaves for each display. The connection between the displays and the wires was achieved through the use of male and female headers (Figure 3-b). This enabled us to replace the displays easily in case they were broken. The displays switch their state in about 40 seconds.

3 DISCUSSION AND CONCLUSION

Wearable designs with NFC technology have already been presented as ways to increase sustainability by achieving energy autonomous devices for data exchange [3, 7] and powering expressive visual changes on wearables [4, 5]. Different from these works, DecoLeaves advances the possibilities of NFC by suggesting NFC as energy autonomous and concealed way of controlling one’s own visual expression. We exemplified the secret movement of the smartphone within the pocket to trigger two NFC readers embedded in the pocket design. We believe that this approach can be extended, for instance, by placing NFC readers on the sleeves. With such a design, the wearer can hover her hand over her pocket where the phone is placed to secretly activate visual changes on the garment. In this regard, our work opens a design space where concealed input opportunities can be explored by using NFC technology in a battery-free interactive fashion.

Furthermore, by exploring NFC-based secretive interactions, the researchers can create sustainable alternatives for social wearables to influence collocated social interactions by designing *magical* input-output interplay on interactive fashion [2]. For example, we foresee that concealed interactions with NFC can be implemented in self-expressive use cases, such as symbolically communicating a "stay away" message as suggested in Spider dress [12], without needing a battery on the garment. Indeed, in particular to our design, we are curious about what kind of other use-cases our concept can enable in real-life. For this purpose, in our future work, we aim to conduct an exploratory user study where we ask people to try out our interactive fashion in different contexts and collect data about when and why the wearers want to change the visual expressions, as well as capturing the value of concealed interaction to achieve this.

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